

Jeevan U Gowda

📍 Bengaluru, India ✉ jeevanu345@gmail.com ☎ +91 63627 24307 in [jeevanu345](#) 🌐 [jeevanu345](#)

Education

Bachelor of Engineering in Information Science & Engineering

Aug 2022 – Jun 2026

B.M.S College of Engineering, VTU – Bengaluru

- Expected Graduation: July 2026 — Cumulative GPA: 8.4/10.0
- **Relevant Coursework:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Networks, Software Engineering, Machine Learning
- **Extracurriculars:** Vocalist, Guitarist, Football

12th Grade (Karnataka State Board) – 90%

Apr 2021 – Jun 2022

10th Grade (CBSE Board) – 90%

2019 – 2020

Experience

Software Development Engineer Intern – Brillio, Bengaluru

Mar 2026 – Present

Projects

RaftKV: Raft-Replicated Key-Value Store

- Built a Raft-replicated key-value store in Rust with a Redis-compatible (RESP2/RESP3) wire protocol and gRPC peer-to-peer transport. Consensus core is a pure state machine emitting side-effects as typed **Action** values, enabling deterministic simulation testing.
- Ships with a segmented durable log, sled-backed KV state machine, linearizable reads via read-index protocol, a Wing-and-Gong linearizability checker, and a Helm chart for Kubernetes deployment.
- **Tech Stack:** Rust, gRPC (tonic), RESP2/RESP3, sled, Docker, Kubernetes (Helm), Grafana

Code-Repair Agent with Execution-Grounded RL on SWE-Bench

- Trained an LLM agent to autonomously resolve real GitHub issues from SWE-bench Lite/Verified using GRPO. Agent follows a ReAct loop with tools (bash, file read/write/edit, grep, repo map, test run) inside an ephemeral, resource-capped, network-disabled Docker sandbox.
- Reward is binary and execution-grounded: FAIL_TO_PASS tests must pass and PASS_TO_PASS tests must not regress. Rollouts are fully reproducible via recorded seed, image digest, model SHA, and repo commit.
- **Tech Stack:** Python, TRL (GRPO), vLLM, Docker, Ray, PostgreSQL, Redis, Terraform, Prometheus, W&B

NCERT RAG Bot: AI-Powered NCERT Tutor

- Built an intelligent AI tutor using a RAG pipeline for conversational Q&A from NCERT Class 10 PDFs with local LLM integration, smart text chunking, vector search, and memory-based context retention.
- **Tech Stack:** Python, Streamlit, Ollama (phi3-fast), Chroma Vector DB, LangChain, PyPDF, FAISS

Instant Canteen: Smart Campus Food Ordering & Payment System

- Built a full-stack web app streamlining campus canteen operations — enabling students to pre-order meals and make instant UPI-based QR payments, while providing vendors a real-time dashboard for order and menu management.
- **Tech Stack:** React, TypeScript, Tailwind CSS, Vite, Supabase, Netlify

TapCart: NFC-Based Smart Retail System

- Designed an IoT-integrated retail platform enabling “tap-to-shop” purchases and real-time inventory sync across devices.
- **Tech Stack:** NFC, IoT, React Native, Node.js, Express.js, Cloud APIs

COVID-19 Genome Mutation Visualization Tool

- Built a visualization tool to compare genome sequences of COVID-19 variants and visualize protein structure mutations via ChimeraX.
- **Tech Stack:** Python, Biopython, ChimeraX, Matplotlib, Phylogenetic Visualization

Research Papers

TapCart: NFC-Based Payment Interface

Apr 2025

- Proposed secure NFC-based merchant payment architecture for contactless transactions without extra hardware.

Technical Skills

Languages: C++, Python, C, Java, JavaScript, Rust, HTML/CSS

Frameworks & Tools: React.js, Node.js, Express.js, LangChain, Tailwind CSS, MySQL, MongoDB, Jupyter Notebook, Framer Motion

AI/ML: RAG Pipelines, Knowledge Graphs, Vector Databases (Chroma, FAISS), LLM Integration, GRPO/RL, Scikit-learn,

Pandas, NumPy, Matplotlib, Seaborn

Cloud/DevOps: AWS (EC2, S3, IAM), Docker, Kubernetes, Terraform, Linux, Networking basics (DNS, HTTP, TCP/IP)

About Me

- Responsible, organized, and hardworking individual passionate about technology, innovation, and problem-solving.
- **Soft Skills:** Communication, Critical Thinking, Analytical Problem Solving — **Languages:** English (Fluent), Hindi (Fluent), Kannada (Native)